Dr. FILOTHEOU, Alexandros

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github.com/li9i · google.scholar · Portfolio · References

Robotics Engineer with eight years designing and deploying applications including autonomous navigation, state estimation, and multi-robot orchestration on ground and aerial platforms in real-world environments. Proven track record leading complex multi-partner integrations from architecture to implementation and field deployment across countries. Backed by IEEE publications and single-authored papers in IROS'24 and R&AS.

Skills

English	Native / Fluent (IELTS 8.5 - C2 Proficiency)
Languages	`C/C++`, `Python`, `shell`, `MATLAB/Octave`
Robotics/OS	`Linux`, `ROS/ROS 2`
Tools/Frameworks	`git`, `Docker`, Behavior Trees, Gazebo, `Eigen`, `CI/CD`, `Qt/Tkinter`
Sensors/Hardware	Ouster 3D/Hokuyo 2D LiDARs, SICK Visionary-T, Astra, Realsense, ZED2, Impinj R420
SLAM, Navigation/Control	`amcl`, `slam_toolbox`, `rtabmap`, `openkarto`, `teb_local_planner`, `nav2`, MPC

Highlights

- Deployed €10M robotics construction project RoBétArmé after 2 years of continuous integration between 12 partners at EDF's Hermillon hydroelectric powerplant in cooperation with Bouygues Construction
- Designed and deployed full-stack autonomous social robot (NLP, Navigation, SLAM, RFID, HRI+GUI) at AMTh museum, engaging 2,000+ visitors annually since 2023
- Identified bottlenecks → reduced deployment time >10x across fleet of additive manufacturing robots
- Engineered 50% boost in localisation accuracy for fleet of RFID-inventorying robots → achieved `cm` accuracy
- Resolved cross-machine `ROS/ROS 2` interoperability for entire 12-partner consortium→unblocked all teams
- One of only eighteen authors of single-authored publications presented at `IEEE IROS'24` (1,587 total)

Representative work

Demos	State Estimation	Lidar Odometry	Publications
Integration [1] · [2]	Video	Video	Estimation [Local] · [Global]
SLAM/Navigation	Paper	Paper	Localisation/Control (solo)
Museum Robot	`Code`	`Code`	Multi-robot MPC
Control is Fun			Navigation Survey

Professional Experience

Senior Robotics Software Engineer · Intermodalics, Leuven BE — Feb 2026 – Present
- At KION Group AG: Bridging VDA5050 fleet manager with `ROS 2` mission behaviours within global team developing mass-production industrial autonomous forklift AMRs for warehouses

Integration and Deployment Engineer · CERTH/ITI, Thessaloniki GR — Sep 2023 – Jan 2026
- Architected and implemented software integration for 12-partner EU Horizon project RoBétArmé: Docker infrastructure, CI/CD pipelines, and cross-machine `ROS/ROS 2` interoperability via `Zenoh` & `ros1_bridge`
- Led Task Planning & Behaviour Tree Orchestration for concrete- and metal-additive manufacturing robots
- Defined development, integration, and deployment principles for all 12 SW teams and >50 Dockerised `ROS/ROS 2` packages; ensured code quality via googletest and cpplint

Robotics & Control Engineer · ECE Dept., Aristotle University of Thessaloniki GR — Sep 2018 – Mar 2023
- Sole developer of all robotics SW for R&D projects RELIEF & CultureId: autonomous ground (Turtlebot 2, Robotnik RB1) and aerial platforms for RFID inventorying; social robot for interactive museum engagement
- Delivered & deployed 2D/3D SLAM and autonomous navigation for warehouses, libraries, and museums
- Developed and integrated codebases producing 18+ publications in top-tier IEEE journals & conferences

Computer Vision Engineer · PANDORA Robotics, Thessaloniki GR — Oct 2013 – Jul 2014
- Gained 2$^{\text{nd}}$ place in 2015 RoboCup Rescue competition developing Kinect-based C++ wall-hole detection

Open Source Contributions to `ROS 2`

- `hitch_estimation_apriltag_array` · Estimation of angle between vehicle and hitched trailer
- `pointcloud_to_ply` · Transformation of pointcloud into mesh and storage in `.ply` or `.obj` format

Educational Background

Doctorate · Electrical & Computer Engineering · Aristotle University of Thessaloniki — Sep 2018 – Jun 2023
Master of Science · Systems, Control, and Robotics · KTH Royal Inst. of Technology — Sep 2015 – Jun 2017
Thesis: Decentralised MPC for multi-robot constrained navigation [ECC'18, IJC'18]
Diploma · Electrical & Computer Engineering · Aristotle University of Thessaloniki — Sep 2005 – Jul 2013